

Discussion 9: Vector differential calculus – Solutions

Names:

Section:

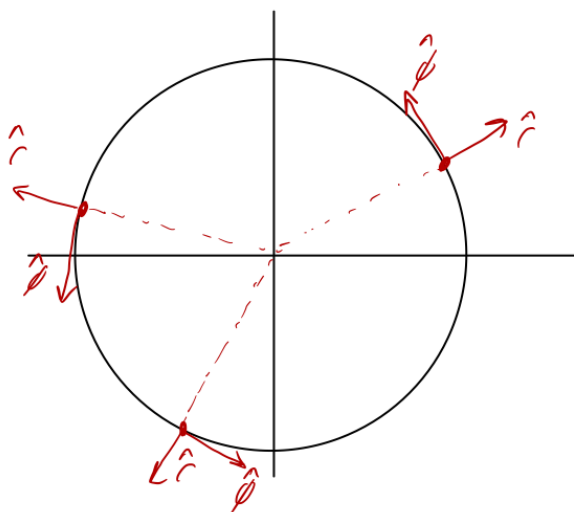
In the first half of the course, we learned that vectors are characterized by their transformations under changes of coordinates. In this week's lecture, you saw how the operations of multivariable calculus can be performed on vectors. This discussion will explore some beautiful connections between these two perspectives. As a bonus, this set of tools is also practically useful – it gives you a quick and easy way to convert between the components of unit vectors in different coordinate systems.

Part 1: Vectors as derivatives

As a warm-up, we will start in 2 dimensions, and explore the relationship between vectors in Cartesian coordinates and polar coordinates. The relations between these are

$$\begin{cases} x = r \cos \phi \\ y = r \sin \phi \end{cases} \iff \begin{cases} r = \sqrt{x^2 + y^2} \\ \phi = \tan^{-1}(y/x) \end{cases}.$$

- 1.1. Draw an xy -plane and a unit circle in that plane centered at the origin. Pick three points on that circle and, attached to those points, draw vectors in the $\hat{\mathbf{r}}$ and $\hat{\phi}$ directions. For this exercise, do this purely geometrically without any calculation: $\hat{\mathbf{r}}$ points outward along radii of the unit circle, in the direction of increasing r , and $\hat{\phi}$ is tangent to the circle, in the direction of increasing ϕ . You should find that $\hat{\mathbf{r}}$ and $\hat{\phi}$ are always perpendicular, but point in different directions relative to the Cartesian unit vectors $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ at different locations on the circle; make sure to check with your instructor if there is any confusion on this point!



- 1.2. To do calculations, it's helpful to have an algebraic formula for the unit vectors in addition to a geometric picture. The polar unit vectors are given in terms of the Cartesian ones by

$$\hat{\mathbf{r}} = \frac{x}{\sqrt{x^2 + y^2}}\hat{\mathbf{x}} + \frac{y}{\sqrt{x^2 + y^2}}\hat{\mathbf{y}}, \quad \hat{\phi} = \frac{-y}{\sqrt{x^2 + y^2}}\hat{\mathbf{x}} + \frac{x}{\sqrt{x^2 + y^2}}\hat{\mathbf{y}}. \quad (1)$$

Compute $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}$, $\hat{\phi} \cdot \hat{\phi}$, and $\hat{\mathbf{r}} \cdot \hat{\phi}$ to show that these unit vectors are orthonormal ($\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\phi} \cdot \hat{\phi} = 1$, $\hat{\mathbf{r}} \cdot \hat{\phi} = 0$). Check that Eq. (1) makes sense by drawing the polar unit vectors at the Cartesian points (1,0) and (0,1), as well as one of the points you drew on the circle in problem 1.1.

$$\begin{aligned} \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} &= \frac{x^2}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{x}}) + \frac{2xy}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{y}}) + \frac{y^2}{x^2 + y^2}(\hat{\mathbf{y}} \cdot \hat{\mathbf{y}}) = \frac{x^2 + y^2}{x^2 + y^2} = 1 \\ \hat{\phi} \cdot \hat{\phi} &= \frac{y^2}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{x}}) - \frac{2xy}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{y}}) + \frac{x^2}{x^2 + y^2}(\hat{\mathbf{y}} \cdot \hat{\mathbf{y}}) = \frac{x^2 + y^2}{x^2 + y^2} = 1 \\ \hat{\mathbf{r}} \cdot \hat{\phi} &= \frac{-xy}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{x}}) + \frac{x^2 - y^2}{x^2 + y^2}(\hat{\mathbf{x}} \cdot \hat{\mathbf{y}}) + \frac{xy}{x^2 + y^2}(\hat{\mathbf{y}} \cdot \hat{\mathbf{y}}) = -\frac{xy}{x^2 + y^2} + \frac{xy}{x^2 + y^2} = 0. \end{aligned}$$

At (1,0), we have $\hat{\mathbf{r}} = \hat{\mathbf{x}}$ and $\hat{\phi} = \hat{\mathbf{y}}$, which makes sense because the radius is along the x -axis and the tangent to the circle is in the y direction. Likewise at (0,1) we have $\hat{\mathbf{r}} = \hat{\mathbf{y}}$ and $\hat{\phi} = -\hat{\mathbf{x}}$, where the minus sign in the tangent comes because $\hat{\phi}$ always points counterclockwise around the circle.

- 1.3. Solve Eq. (1) for $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ in terms of $\hat{\mathbf{r}}$ and $\hat{\phi}$. Make sure you express the coefficients purely in terms of the polar variables r and ϕ ; in other words, x and y should not appear in your answer.

We can eliminate $\hat{\mathbf{y}}$ by multiplying the $\hat{\mathbf{r}}$ equation by x and subtracting the $\hat{\phi}$ equation multiplied by y :

$$x\hat{\mathbf{r}} - y\hat{\phi} = \frac{x^2 + y^2}{\sqrt{x^2 + y^2}}\hat{\mathbf{x}}.$$

Since $\sqrt{x^2 + y^2} = r$, the right-hand side is $r\hat{\mathbf{x}}$, so we have

$$\hat{\mathbf{x}} = \frac{x}{r}\hat{\mathbf{r}} - \frac{y}{r}\hat{\phi} = \cos\phi\hat{\mathbf{r}} - \sin\phi\hat{\phi}.$$

Similarly, we can eliminate $\hat{\mathbf{x}}$ by considering $y\hat{\mathbf{r}} + x\hat{\phi}$:

$$y\hat{\mathbf{r}} + x\hat{\phi} = \frac{y^2 + x^2}{\sqrt{x^2 + y^2}}\hat{\mathbf{y}} \implies \hat{\mathbf{y}} = \frac{y}{r}\hat{\mathbf{r}} + \frac{x}{r}\hat{\phi} = \sin\phi\hat{\mathbf{r}} + \cos\phi\hat{\phi}.$$

- 1.4. The definition of a unit vector is one that points in the direction of increasing coordinate value. For example, $\hat{\mathbf{x}}$ points in the direction of increasing x (i.e. along the x -axis), and $\hat{\mathbf{r}}$ points in the direction of increasing r . But this is exactly what partial derivatives do! For

example, $\partial f/\partial r$ tells us how much a function f is changing in the r direction. It is therefore appropriate to *associate unit vectors with partial derivatives*:

$$\hat{\mathbf{x}} \leftrightarrow \frac{\partial}{\partial x}, \quad \hat{\mathbf{y}} \leftrightarrow \frac{\partial}{\partial y} \quad (2)$$

$$\alpha_r \hat{\mathbf{r}} \leftrightarrow \frac{\partial}{\partial r}, \quad \alpha_\phi \hat{\boldsymbol{\phi}} \leftrightarrow \frac{\partial}{\partial \phi} \quad (3)$$

However, the vectors obtained in this way are not necessarily normalized, which is why we have included factors α_r and α_ϕ .

Now let's put this to use: derive Eq. (1) using the chain rule, and then normalizing the resulting vector to obtain a unit vector. For example, to calculate $\hat{\mathbf{r}}$, use

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y},$$

replace $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$ with $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, and normalize the right-hand side. You should find $\alpha_r = 1$ and $\alpha_\phi = r$.

Since $\partial x/\partial r = \cos \phi$ and $\partial y/\partial r = \sin \phi$, we have

$$\begin{aligned} \frac{\partial}{\partial r} &= \cos \phi \frac{\partial}{\partial x} + \sin \phi \frac{\partial}{\partial y} \\ \leftrightarrow \alpha_r \hat{\mathbf{r}} &= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}} = \frac{x}{r} \hat{\mathbf{x}} + \frac{y}{r} \hat{\mathbf{y}} = \frac{x}{\sqrt{x^2 + y^2}} \hat{\mathbf{x}} + \frac{y}{\sqrt{x^2 + y^2}} \hat{\mathbf{y}}. \end{aligned}$$

In the second line we replaced the partial derivatives with unit vectors. Note that since the right-hand side is already normalized, we have $\alpha_r = 1$. Similarly, $\partial x/\partial \phi = -r \sin \phi$ and $\partial y/\partial \phi = r \cos \phi$, so

$$\begin{aligned} \frac{\partial}{\partial \phi} &= -r \sin \phi \frac{\partial}{\partial x} + r \cos \phi \frac{\partial}{\partial y} \\ \leftrightarrow \alpha_\phi \hat{\boldsymbol{\phi}} &= -r \sin \phi \hat{\mathbf{x}} + r \cos \phi \hat{\mathbf{y}} = -y \hat{\mathbf{x}} + x \hat{\mathbf{y}}. \end{aligned}$$

At this point, we take the dot product of both sides to find $\alpha_\phi^2 = x^2 + y^2$, so $\alpha_\phi = \sqrt{x^2 + y^2} = r$. (Notice that by dimensional analysis, α_ϕ must be proportional to a length, since $\partial/\partial r$ has dimensions of length^{-1} and $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ must have the same dimensions in order to be added together in linear combinations.) Dividing by r to normalize, we obtain the expression in Eq. (1).

- 1.5. Using the same method as problem 1.4, derive your result in problem 1.3 for the Cartesian unit vectors in terms of the polar unit vectors. You will find the following derivative useful:

$$\frac{d}{dw} \tan^{-1}(w) = \frac{1}{1 + w^2}.$$

To obtain the Cartesian unit vectors, we use $\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi}$ and likewise for $\frac{\partial}{\partial y}$. The partial derivatives we need are

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{r} = \cos \phi, \quad \frac{\partial r}{\partial y} = \frac{2y}{2\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin \phi,$$

$$\frac{\partial \phi}{\partial x} = \frac{-y/x^2}{1 + (y/x)^2} = -\frac{y}{x^2 + y^2} = -\frac{\sin \phi}{r}, \quad \frac{\partial \phi}{\partial y} = \frac{1/x}{1 + (y/x)^2} = \frac{x}{x^2 + y^2} = \frac{\cos \phi}{r}.$$

Now following the method of problem 1.4,

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \\ \Leftrightarrow \alpha_x \hat{\mathbf{x}} &= \cos \phi \hat{\mathbf{r}} - \frac{\sin \phi}{r} (\alpha_\phi \hat{\phi}) = \cos \phi \hat{\mathbf{r}} - \sin \phi \hat{\phi}. \end{aligned}$$

This is already normalized, so $\alpha_x = 1$ and we recover the result of problem 1.3. Similarly,

$$\begin{aligned} \frac{\partial}{\partial y} &= \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \\ \Leftrightarrow \alpha_y \hat{\mathbf{y}} &= \sin \phi \hat{\mathbf{r}} + \frac{\cos \phi}{r} (\alpha_\phi \hat{\phi}) = \sin \phi \hat{\mathbf{r}} + \cos \phi \hat{\phi}, \end{aligned}$$

which is also normalized, so $\alpha_y = 1$.

Part 2: The gradient in spherical coordinate systems

Now let's move to 3 dimensions and consider spherical coordinates:

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \iff \begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) \\ \phi = \tan^{-1} \left(\frac{y}{x} \right) \end{cases}.$$

Recall the general definition of the gradient operator you saw in lecture:

$$\nabla \equiv \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}.$$

- 2.1. Using the techniques of Part 1, compute the spherical component unit vectors $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ in terms of the Cartesian coordinates and unit vectors. You can leave the coefficients in terms of r , θ , and ϕ rather than converting everything to Cartesian. You should find the normalization factors

$$\hat{\mathbf{r}} \leftrightarrow \frac{\partial}{\partial r}, \quad \hat{\boldsymbol{\theta}} \leftrightarrow \frac{1}{r} \frac{\partial}{\partial \theta}, \quad \hat{\boldsymbol{\phi}} \leftrightarrow \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

A good check of your results is to plug in $\theta = \pi/2$, which describes points in the xy -plane; the resultant unit vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ should reproduce your results from Part 1, and you should find $\hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}$.

$$\begin{aligned} \frac{\partial}{\partial r} &= \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y} + \frac{\partial z}{\partial r} \frac{\partial}{\partial z} \\ \Leftrightarrow \alpha_r \hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}} = \frac{x}{r} \hat{\mathbf{x}} + \frac{y}{r} \hat{\mathbf{y}} + \frac{z}{r} \hat{\mathbf{z}}. \end{aligned}$$

In this form, it's clear that the right-hand side is normalized, so $\alpha_r = 1$ and we can identify $\hat{\mathbf{r}} \leftrightarrow \partial/\partial r$, thus

$$\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}.$$

Continuing on,

$$\begin{aligned} \frac{\partial}{\partial \theta} &= \frac{\partial x}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \theta} \frac{\partial}{\partial z} \\ \leftrightarrow \alpha_\theta \hat{\boldsymbol{\theta}} &= r \cos \theta \cos \phi \hat{\mathbf{x}} + r \cos \theta \sin \phi \hat{\mathbf{y}} - r \sin \theta \hat{\mathbf{z}}. \end{aligned}$$

Taking the dot product of both sides with itself gives

$$\alpha_\theta^2 = r^2 (\cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \sin^2 \theta) = r^2,$$

so $\alpha_\theta = r$ (implying $\hat{\boldsymbol{\theta}} \leftrightarrow \frac{1}{r} \frac{\partial}{\partial \theta}$) and

$$\hat{\boldsymbol{\theta}} = \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}.$$

Finally,

$$\begin{aligned} \frac{\partial}{\partial \phi} &= \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z} \\ \leftrightarrow \alpha_\phi \hat{\boldsymbol{\phi}} &= -r \sin \theta \sin \phi \hat{\mathbf{x}} + r \sin \theta \cos \phi \hat{\mathbf{y}} = r \sin \theta (-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}). \end{aligned}$$

This shows that $\alpha_\phi \hat{\boldsymbol{\phi}}$ in spherical coordinates is $r \sin \theta$ times the polar $\hat{\boldsymbol{\phi}}$ unit vector from Part 1. Thus $\alpha_\phi = r \sin \theta$ (and $\hat{\boldsymbol{\phi}} \leftrightarrow \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$), and

$$\hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}.$$

Plugging in $\theta = \pi/2$ gives $\sin \theta = 1$, $\cos \theta = 0$, so $\hat{\mathbf{r}} = \frac{x}{r} \hat{\mathbf{x}} + \frac{y}{r} \hat{\mathbf{y}}$ and $\hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}$, exactly as found in Part 1, and $\hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}$ since the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components vanish.

- 2.2. The gradient has a very nice expression in index notation. Let the unit vectors be $\mathbf{e}^i$, with $i = 1, 2, 3$ (so $\mathbf{e}^1 = \hat{\mathbf{x}}$, $\mathbf{e}^2 = \hat{\mathbf{y}}$, and $\mathbf{e}^3 = \hat{\mathbf{z}}$). Similarly, let the Cartesian coordinates be x^i and write the associated partial derivatives as $\partial_i \equiv \partial/\partial x^i$. Then we can write

$$\nabla \equiv \mathbf{e}^i \partial_i \tag{4}$$

where we have used the summation convention. Convince yourself that this is equivalent to the definition of the gradient given in class.

Writing out the components,

$$\nabla = \mathbf{e}^1 \partial_1 + \mathbf{e}^2 \partial_2 + \mathbf{e}^3 \partial_3 = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z},$$

as given in class.

- 2.3. Notice that in problem 2.2, each unit vector is paired with its corresponding coordinate derivative, in the sense of problem 1.4. It turns out this works perfectly well for other orthogonal coordinate systems! Using this fact, write down the expression for the gradient in spherical coordinates (summing over the coordinates r , θ , and ϕ instead of x , y , and z), making use of your unit vector-derivative pairs from problem 2.1. You should find exactly the expression for the gradient which you saw in lecture.

Using the unit vector-derivative pairs from problem 2.1:

$$\nabla = \hat{\mathbf{r}} \frac{d}{dr} + \hat{\boldsymbol{\theta}} \left(\frac{1}{r} \frac{\partial}{\partial \theta} \right) + \hat{\boldsymbol{\phi}} \left(\frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right),$$

which is indeed the gradient in spherical coordinates.

Part 3: The gradient is a vector

Recall that in index notation, we often associate a vector with its components: for example, the vector $\mathbf{v} = 2\hat{\mathbf{x}} + 3\hat{\mathbf{y}} - 5\hat{\mathbf{z}}$ has Cartesian components $(2, 3, 5)$, and we can refer to the vector as v^i with $v^1 = 2$, $v^2 = 3$, and $v^3 = 5$.

We can do the same with the gradient, and use Eq. (4) to treat the gradient as a vector with components ∂_i . By the end of this part, you will see why this makes sense: the gradient transforms under rotations in the same way that a (row) vector does, which also explains why the index is naturally downstairs when the coordinates x^i have upstairs indices.

- 3.1. Under a coordinate transformation by a rotation matrix R^i_j , the Cartesian coordinates in the old and new frames are related by $x^i = R^i_j x'^j$. Use the chain rule to show that the components of the gradient transform as

$$\frac{\partial}{\partial x'^j} = R^i_j \frac{\partial}{\partial x^i} \quad (5)$$

Note that this tells us that the gradient contracts with the matrix R the same way that a row vector does (namely $v'_j = v_i R^i_j$), which justifies index notation treating an upper index appearing in the “denominator” of a partial derivative as a lower index.

Note that $x^i = R^i_j x'^j$ is a linear transformation of coordinates, so the partial derivative of (say) x^1 with respect to x'^2 is just its coefficient, which is R^1_2 . We can summarize all possible pairs of x and x' using index notation:

$$\frac{\partial x^i}{\partial x'^j} = R^i_j.$$

The chain rule then gives

$$\frac{\partial}{\partial x'^j} = \frac{\partial x^i}{\partial x'^j} \frac{\partial}{\partial x^i} = R^i_j \frac{\partial}{\partial x^i}.$$

Notice that in the first equality, the summation convention is used with the dummy index i , since the multivariate chain rule involves a sum over the possible partial derivatives with respect to x'^j .

- 3.2. The gradient is special; not all other random differential operators behave like vectors under coordinate rotations! Argue that the operator $\hat{\mathbf{x}}\partial_x + \hat{\mathbf{y}}\partial_y + 2\hat{\mathbf{z}}\partial_z$ does *not* transform in the same way as Eq. (5), and thus it is (despite appearances) *not* a vector.

Consider the coordinate transformation which is a rotation about the x -axis by $\pi/2$: $x = x'$, $y = -z'$, $z = y'$. If the components of this operator in the primed frame are $(\partial_{x'}, \partial_{y'}, 2\partial_{z'})$, then using the chain rule, the transformation rule is

$$(\partial_{x'}, \partial_{y'}, 2\partial_{z'}) \rightarrow \left(\frac{\partial x}{\partial x'} \partial_x, \frac{\partial z}{\partial y'} \partial_z, 2 \frac{\partial y}{\partial z'} \partial_y \right) = (\partial_x, \partial_z, -2\partial_y).$$

On the other hand, rotating the components of the vector by the transformation matrix as in problem 3.1 instructs us to simply swap the y and z components with a minus sign, giving

$$(\partial_{x'}, \partial_{y'}, 2\partial_{z'}) \rightarrow (\partial_x, 2\partial_z, -\partial_y),$$

which is not the same as obtained by the chain rule. So these two transformation rules are inconsistent, and thus the operator is not a vector. (This is an identical argument for why $(dt, 2dx, dy, dz)$ is not a 4-vector, in Morin Section 4.2, which we will cover next week.)